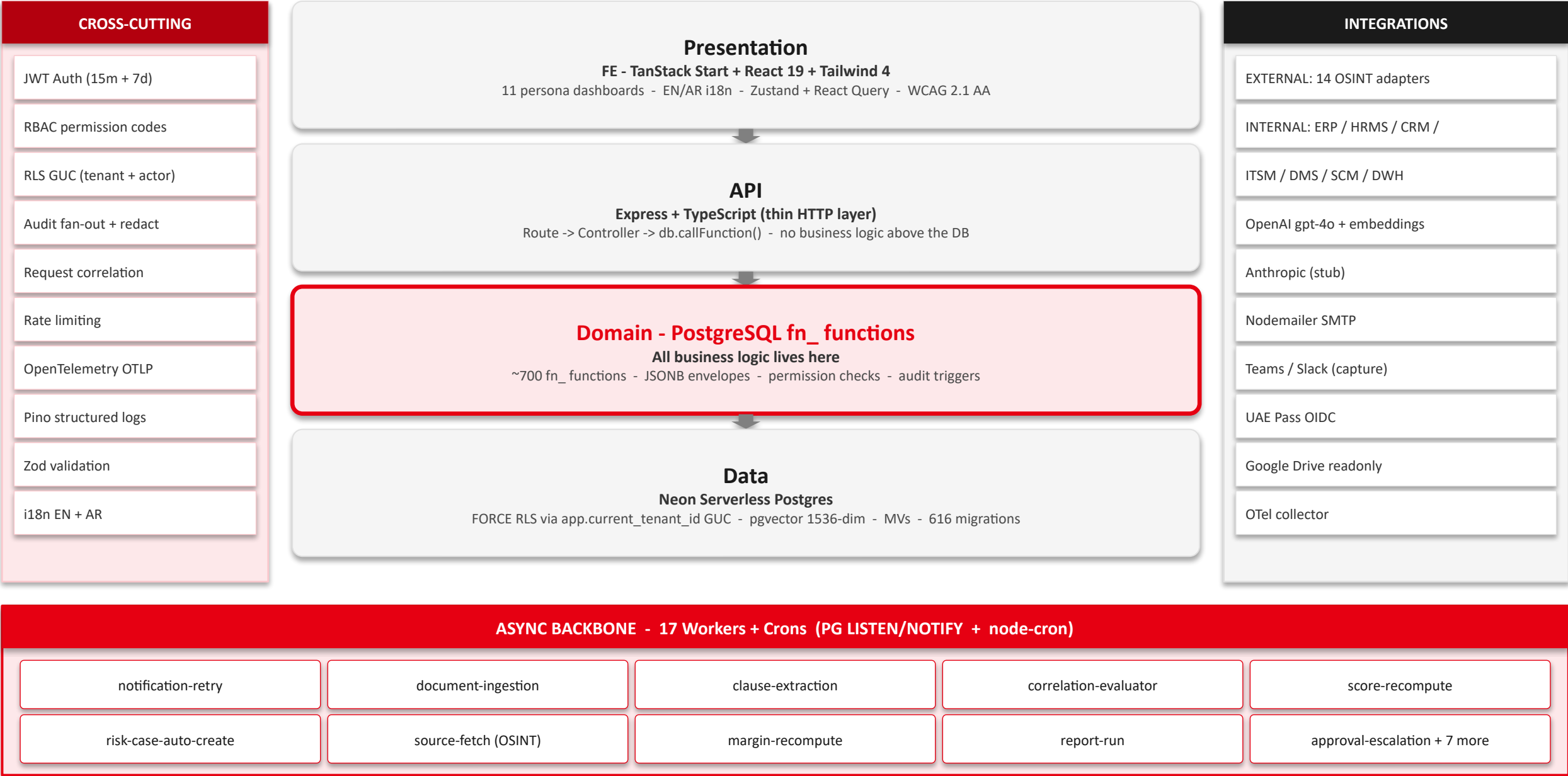


MUSANAD CONTRACTS

CLM + ECIP

Technical Architecture



CONTRACT LIFECYCLE

- > **M1 Contracts**
CRUD, versions, comments, payments
- > **M2 Approvals**
matrix, chains, escalation
- > **M3 Signatures**
token bearer + invitations
- > **M11 Document Ingestion**
PDF + Tesseract OCR + Vision
- > **M12 Clause Extraction**
pgvector semantic search
- > **Templates + Clauses**
library + bulk extract
- > **M21a Budget Burn**
services-contract YTD variance

RISK INTELLIGENCE

- > **M7 OSINT Framework**
14 adapters
- > **M8 Internal Signals**
8 types (SLA / payment / vendor / ...)
- > **M13 Correlation Engine**
30 DSL predicates
- > **M14 5-Dim Risk Scoring**
MaR + AVaR + MV
- > **M19 Risk Cases**
auto-create + escalation
- > **M16 Advisory Drafter**
Mustache + LLM polish
- > **M21b Oil-Trade Margin**
Index-Linked Contracts

AI SURFACES

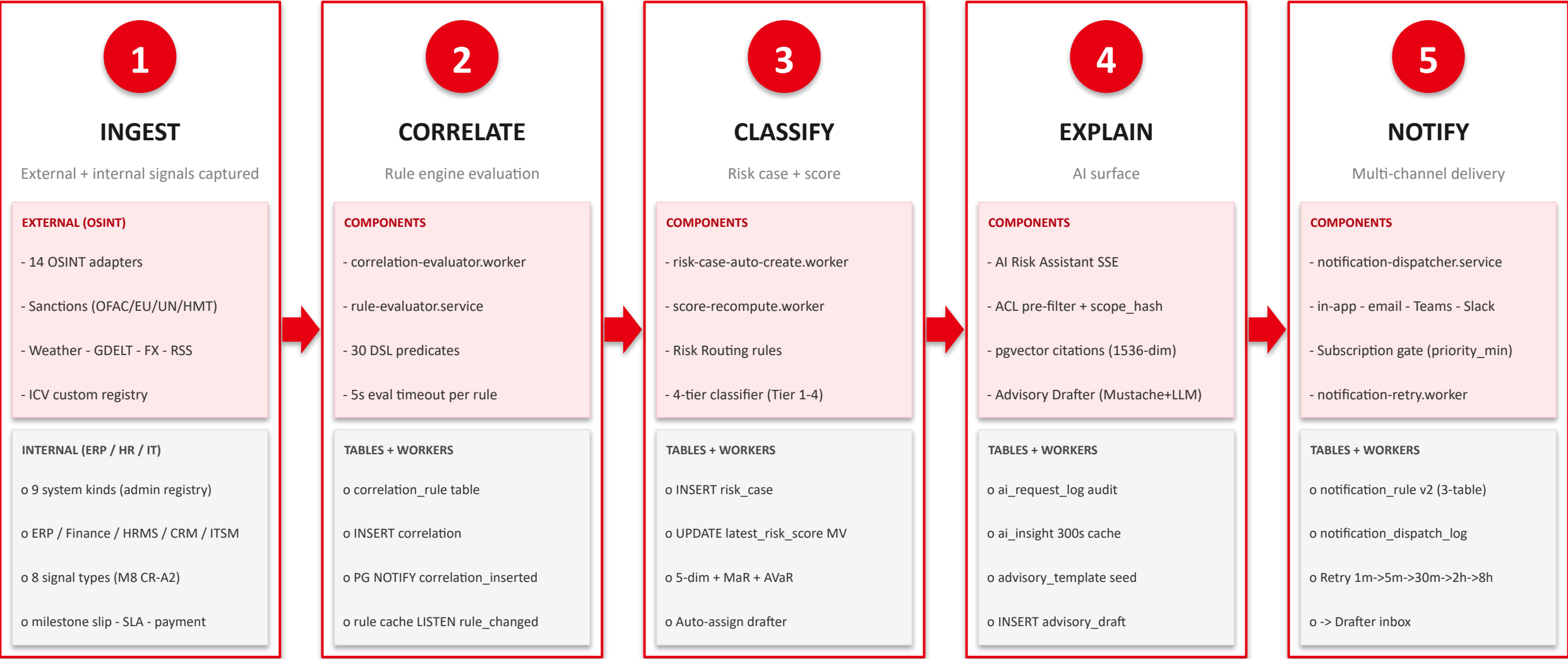
- > **M4 Contract Insights**
10 AI controllers
- > **M15 AI Risk Assistant**
SSE streaming + ACL filter
- > **M15 Persona Dashboards**
Ops / Fin / ESG / Procurement
- > **Drafting Assistant**
version diff + title translate
- > **pgvector Embeddings**
1536-dim, 3 entities
- > **AI Cost Telemetry**
per-call settlement
- > **Prompt Management**
composite_id versioning

NOTIFICATIONS & WORKFLOW

- > **Notification Rule v2**
3-table model + conditions
- > **Dispatch + Retry**
1m -> 5m -> 30m -> 2h -> 8h backoff
- > **Channels**
email - Teams capture - Slack capture - in-app
- > **Subscriptions**
per-user x kind x channel
- > **17 async workers**
cron + LISTEN/NOTIFY
- > **SLA escalations**
approvals - signatures - obligations
- > **Demo time-freeze**
deterministic scenarios

PLATFORM & ADMIN

- > **Multi-tenant chassis**
tenant + product_module_enable
- > **Audit Chain**
tamper-evident verification
- > **Internal Systems Registry**
ERP / Finance / HRMS / CRM / ITSM / DMS / SCM / DWH
- > **M20 Reports & Briefings**
24 ADNOC slugs + PDF / XLSX
- > **M22 Migration + Drive**
Google Drive readonly OAuth
- > **Demo Harness**
8 scenarios + 9-subsystem health
- > **Industry Catalog**
benchmarks + cost components



External + internal signals converge into osint_signal - same pipeline correlates, scores, classifies, explains, and notifies

Every step audited (audit_log + Pino redact) - tenant-scoped (FORCE RLS + GUC) - surfaced in 11 persona dashboards - Teams / Slack are capture-only payloads today